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PROBABILITY

Figure 7.1 Roulette is a game whose outcomes are based entirely on the concept of probability. (credit: modification of work "Roulette wheel" by Håkan Dahlström/Flickr, CC BY 2.0)

Chapter Outline

- 7.1 The Multiplication Rule for Counting
- 7.2 Permutations
- 7.3 Combinations
- 7.4 Tree Diagrams, Tables, and Outcomes
- 7.5 Basic Concepts of Probability
- 7.6 Probability with Permutations and Combinations
- 7.7 What Are the Odds?
- 7.8 The Addition Rule for Probability
- 7.9 Conditional Probability and the Multiplication Rule
- 7.10 The Binomial Distribution
- 7.11 Expected Value



Introduction

Casinos are big business; according to the American Gaming Association, commercial casinos in the United States brought in over \$43 billion in revenue in 2019. Casinos must walk a fine line in order to be profitable. Their customers must lose more money than they win, on average, in order to stay in business. But if the chances of a single customer winning more money than they lose is too small, people will stop coming in the door to play the games.

In this chapter, we'll study the techniques a casino must use to determine how likely it is that a customer will win a particular game, and then how the casino decides how much money a winner will rake in so that the customers are happy, but the casino also turns a profit in the long run. In order to figure out those likelihoods, we have to be able to somehow consider every possible outcome of these games. For example, in a game that involves players receiving 5 cards from a deck of 52, there are 2,598,960 possibilities for each player. We'll start off this chapter by learning how to count those possible outcomes.